Lecture 20 (11/2)

Residual plots

Today:

1) Residual plots to identify interactions and other model deficiencies

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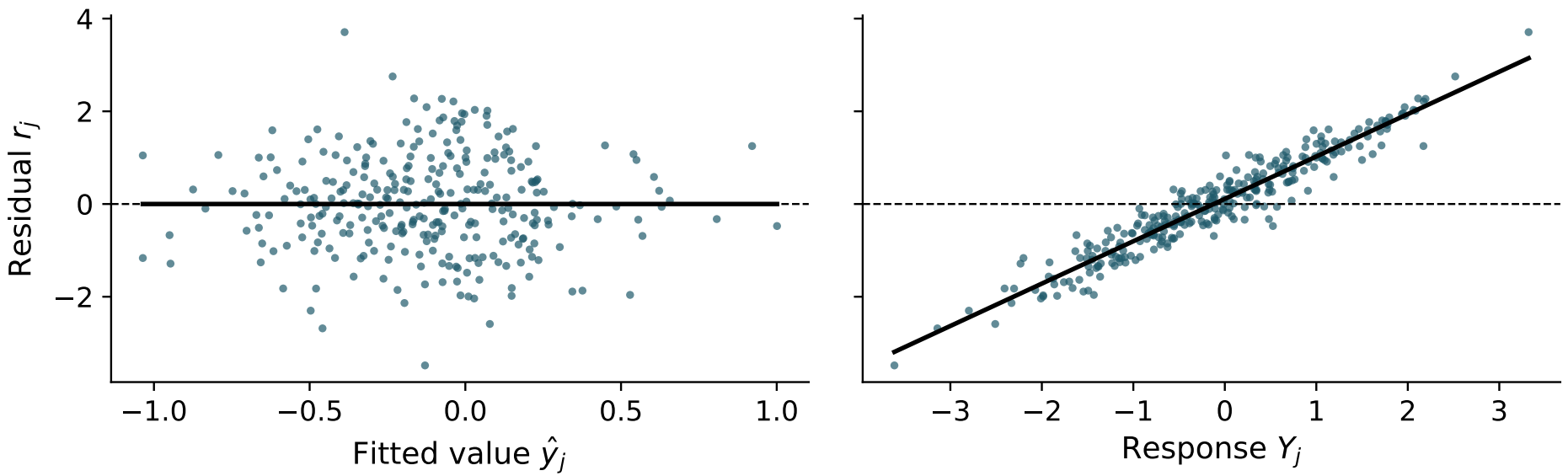
(r_j is computed from the data; ϵ_j in the model cannot be observed.)

If the model is right, r_j is approximately Normal$(0, \sigma_\epsilon^2)$ *whatever the values of the predictors*. A residual plot is meant to visualize the r_j.

Residuals vs. fitted values and vs. Y

Consider

$$Y = 0.5X + \epsilon, \quad \sigma_\epsilon = 1$$



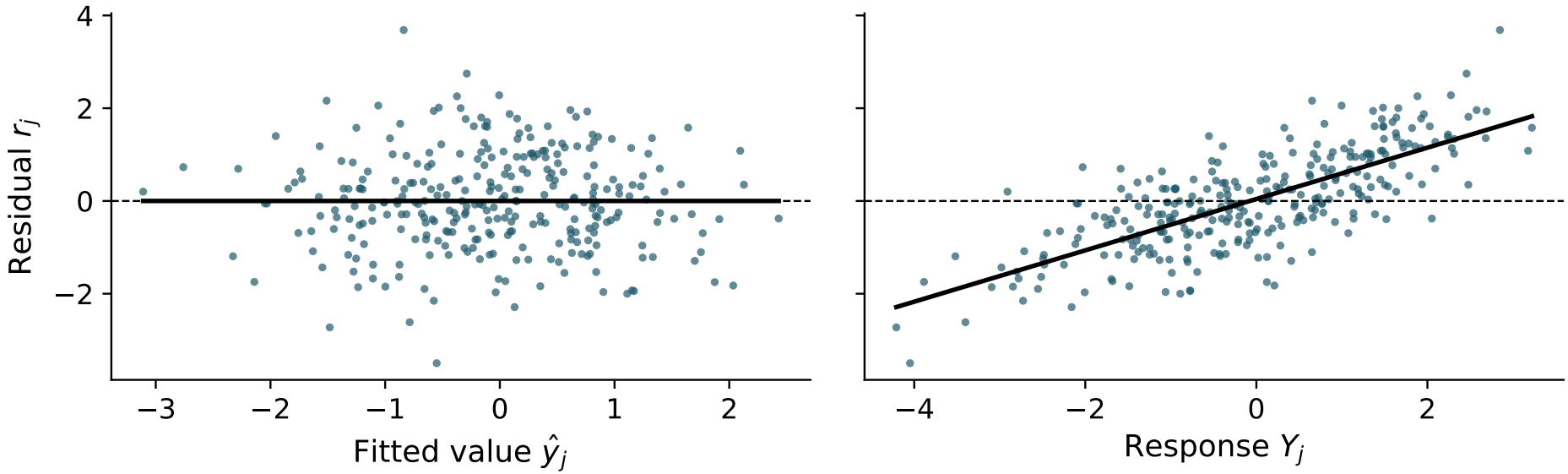
Question

Why is there a trend on the right plot?

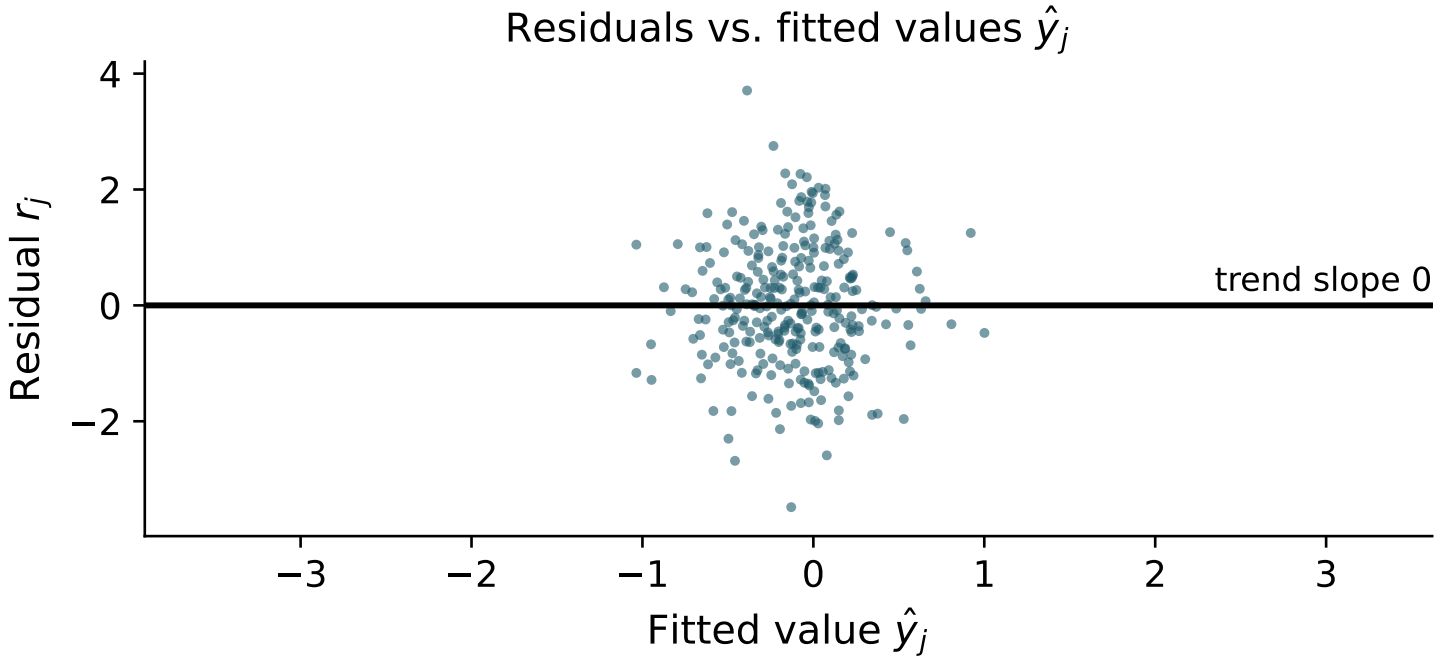
Residuals vs. fitted values and vs. Y: two predictors

Consider

$$Y = 0.2X_1 + 20X_2 + \epsilon, \quad \sigma_\epsilon = 1$$

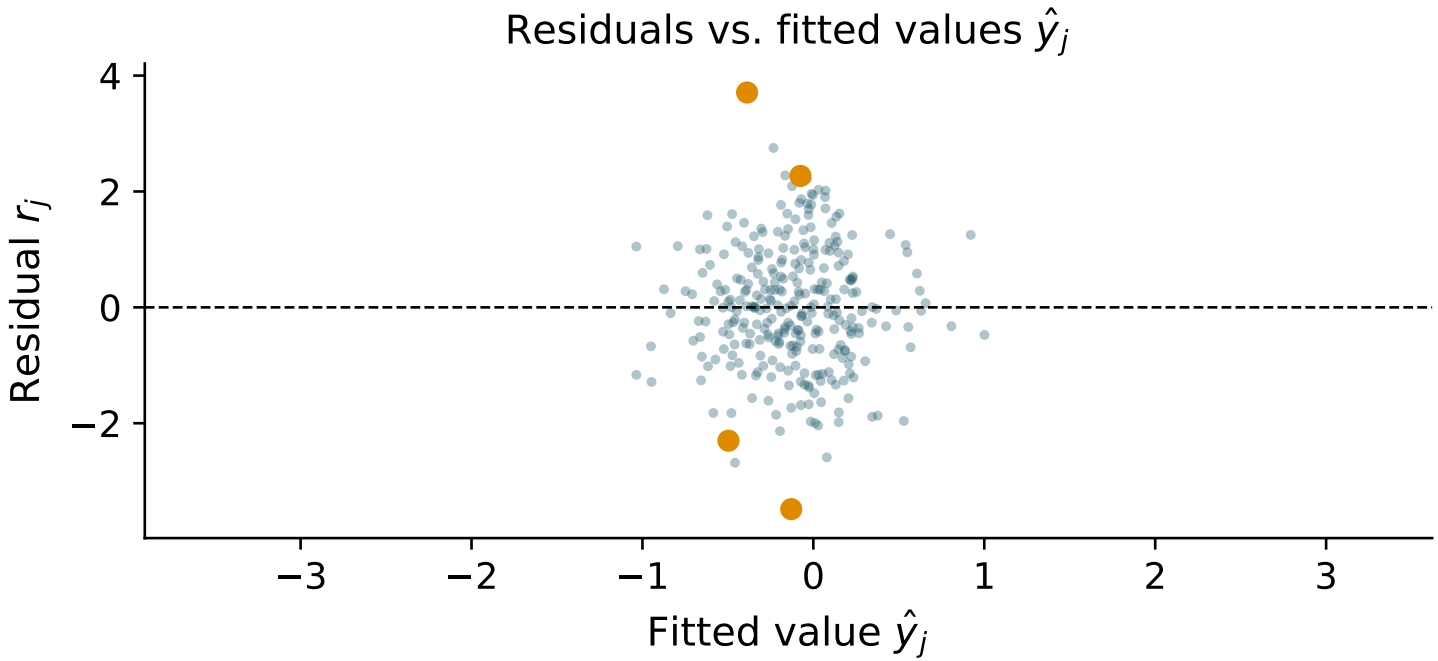


Why plot $\hat{y}_i$ on x axis



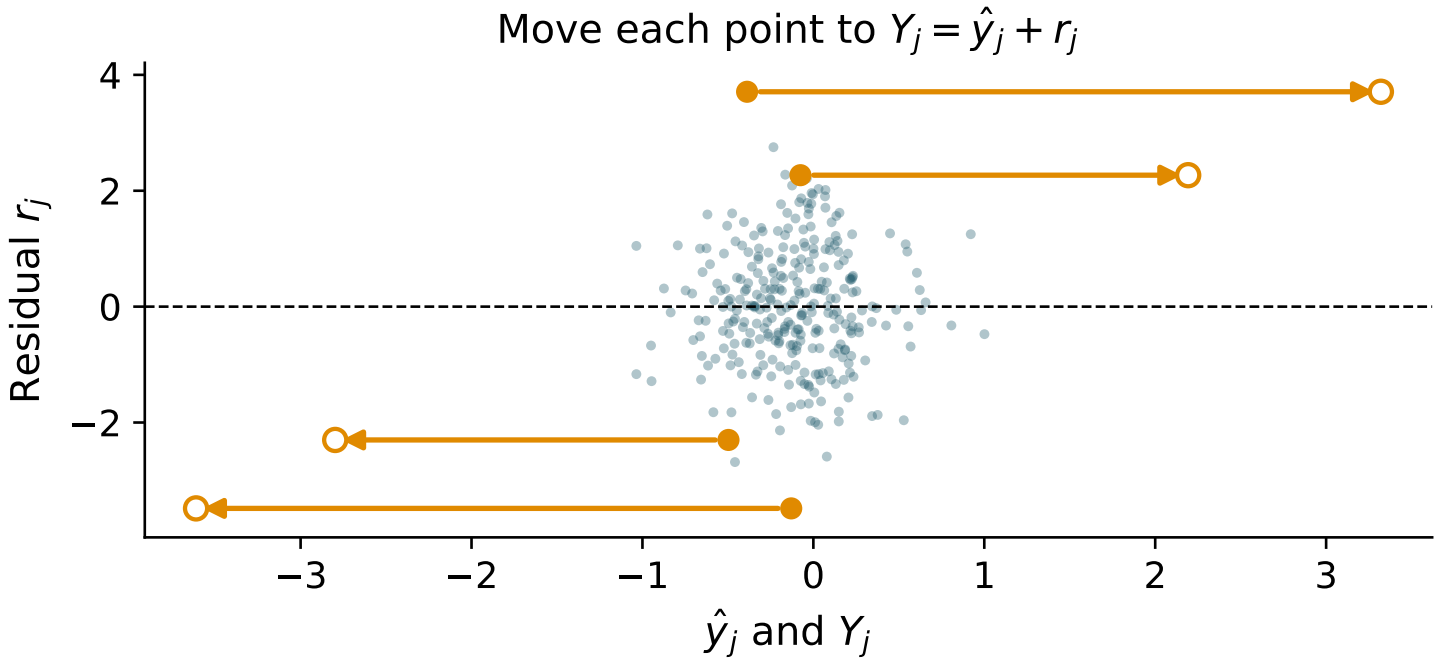
1. If the model is right, r_j has the same distribution whatever the predictors are, so vs. $\hat{y}_j$ we expect a symmetric spread in the vertical direction.

Why plot $\hat{y}_i$ on x axis



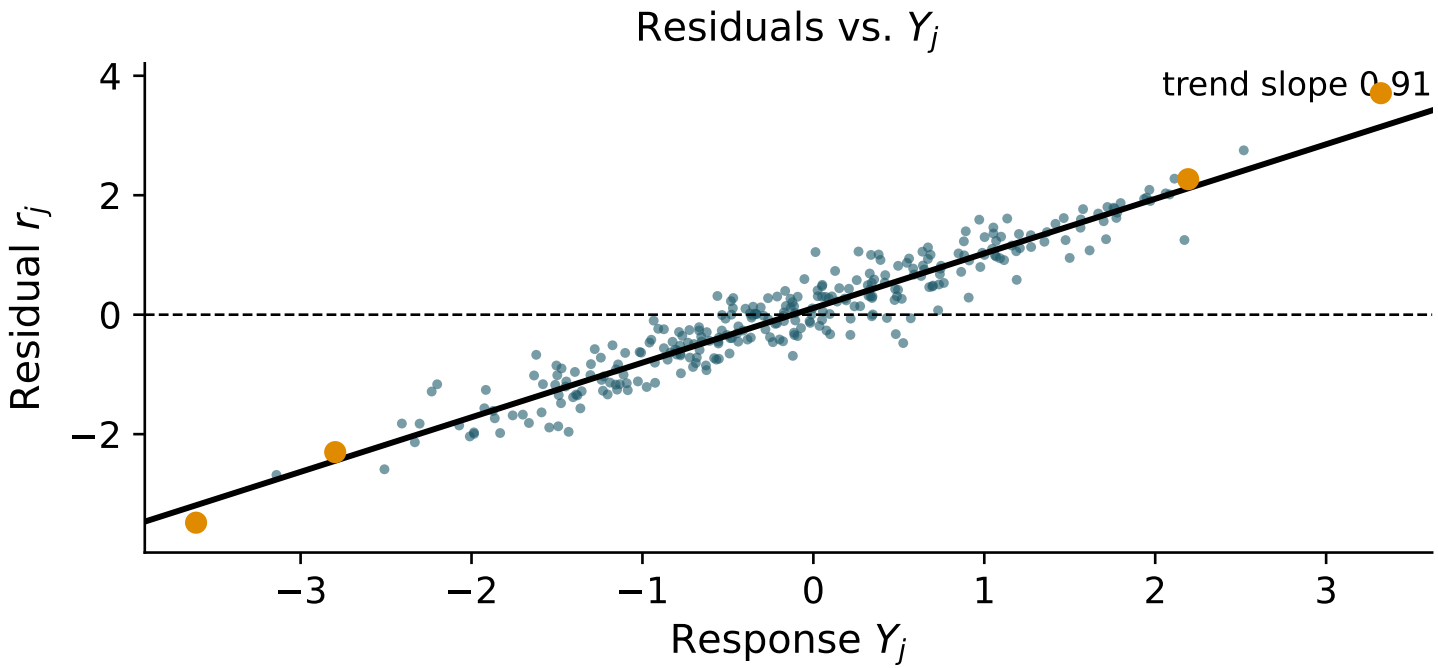
2. Pick out a few points with large residuals.

Why plot $\hat{y}_j$ on x axis



3. Since $Y_j = \hat{y}_j + r_j$, putting Y_j on the horizontal axis shifts each point sideways by the corresponding residual.

Why plot $\hat{y}_i$ on x axis



4. Every point shifts this way, so the cloud tilts even though the model is correct.

Example

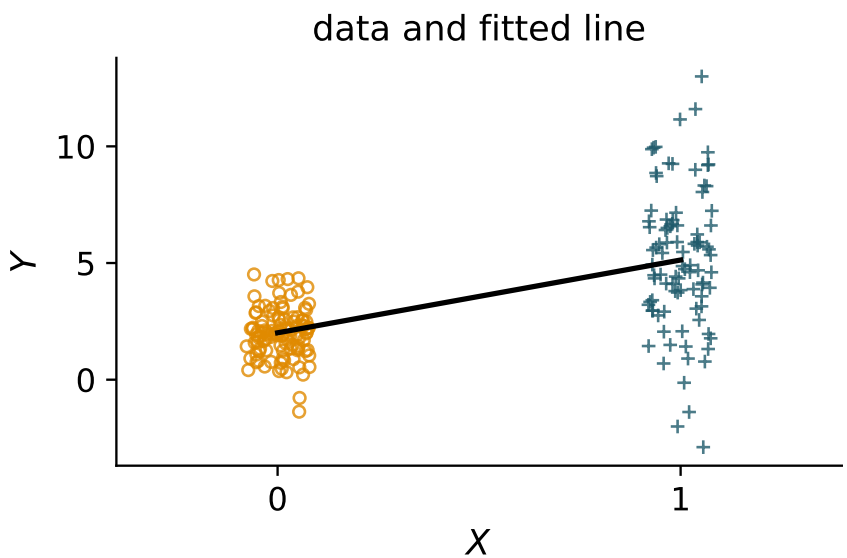
Consider

$$Y = 2 + 3X + \epsilon(1 + 2X)$$

Below is the data with the fitted line.

Question

What will the residual plot look like when the data is fit to a linear regression model?



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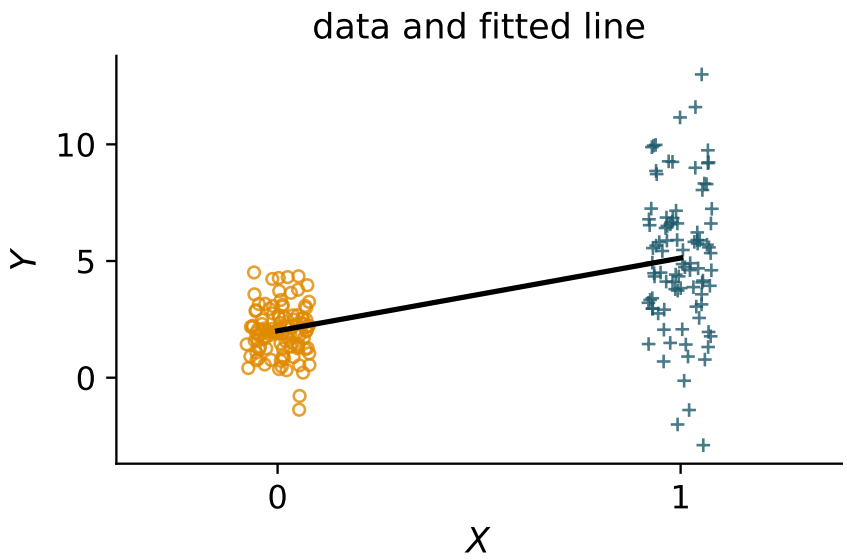
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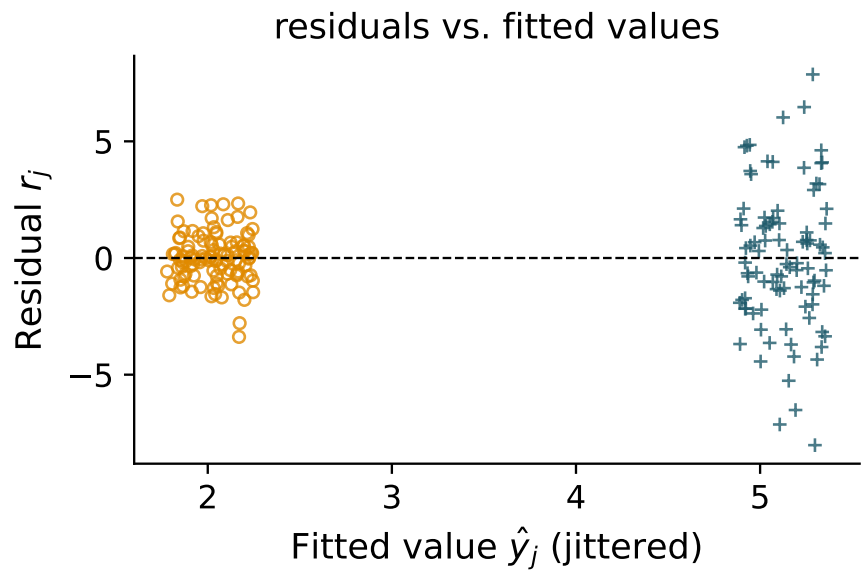
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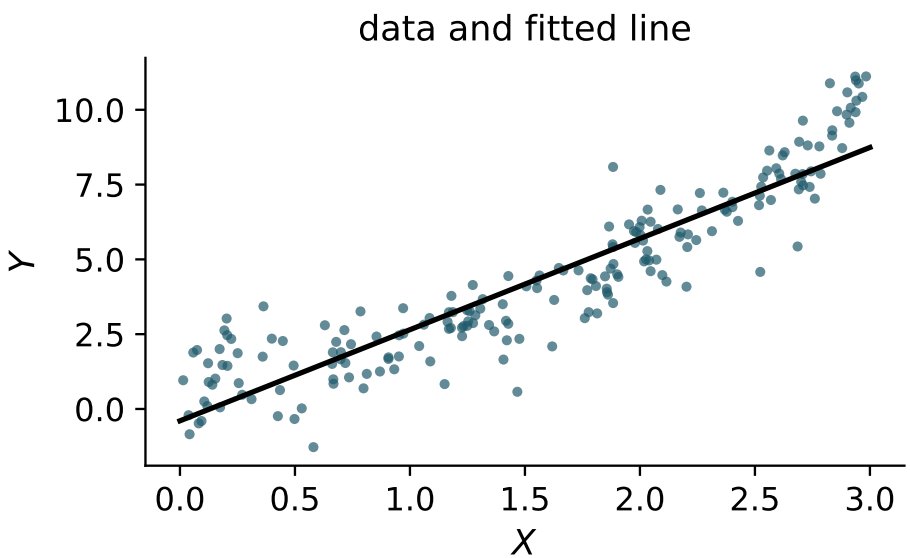
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$$Y = 1 + X^2 + \epsilon$$

where X is between 0 and 3. Below is the data with the fitted line.

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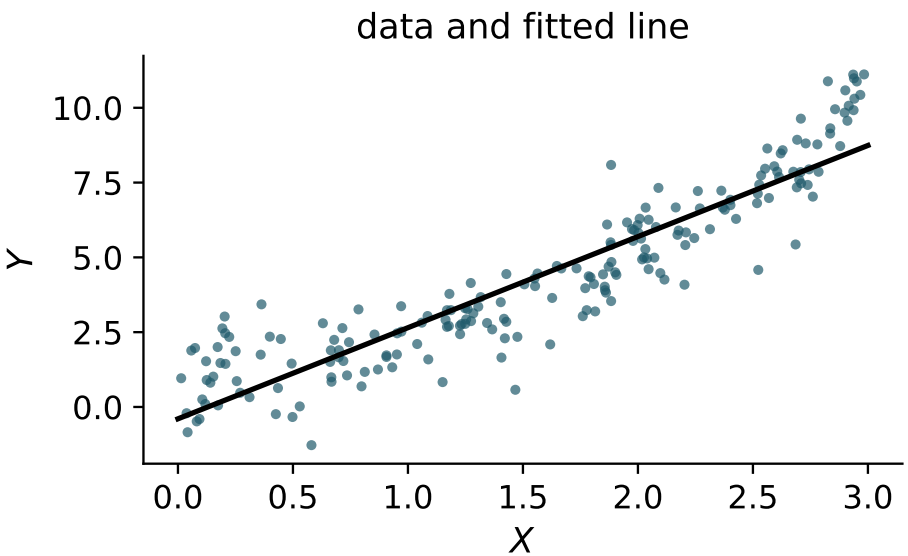
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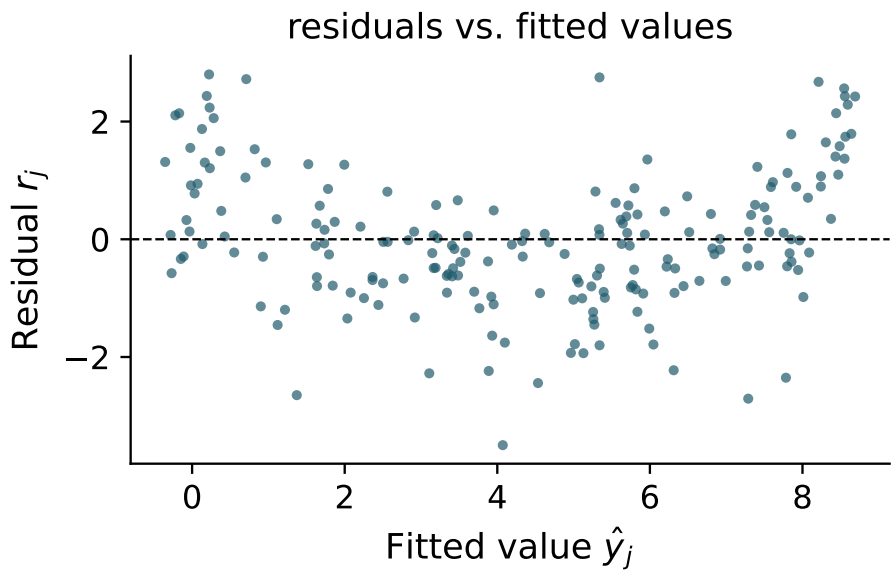
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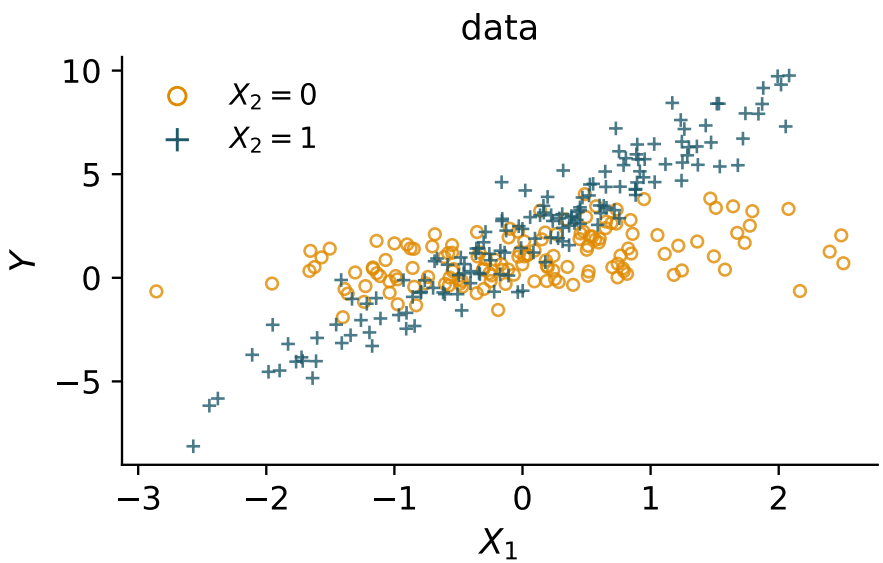
Consider

$$Y = 1 + 0.5X_1 + X_2 + 3X_1X_2 + \epsilon$$

where X_2 is a binary predictor. Below is the data.

Question

Sketch the residual plot when this model is fitted without the interaction term.



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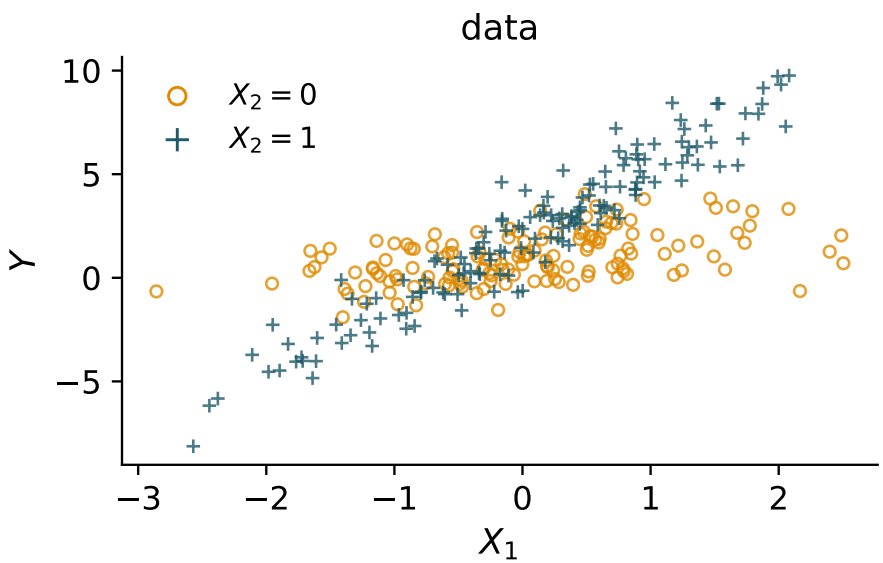
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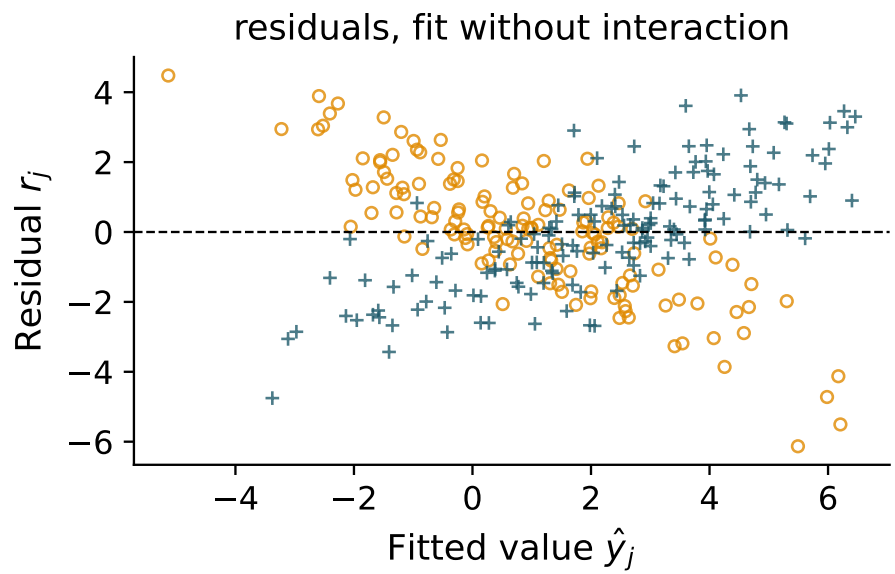
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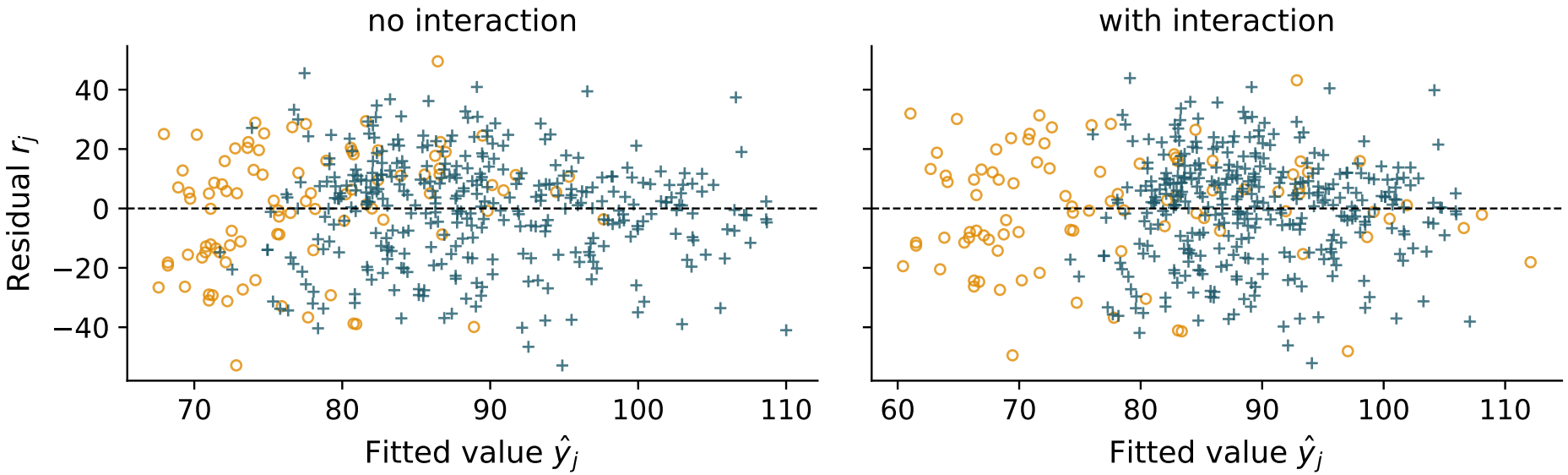
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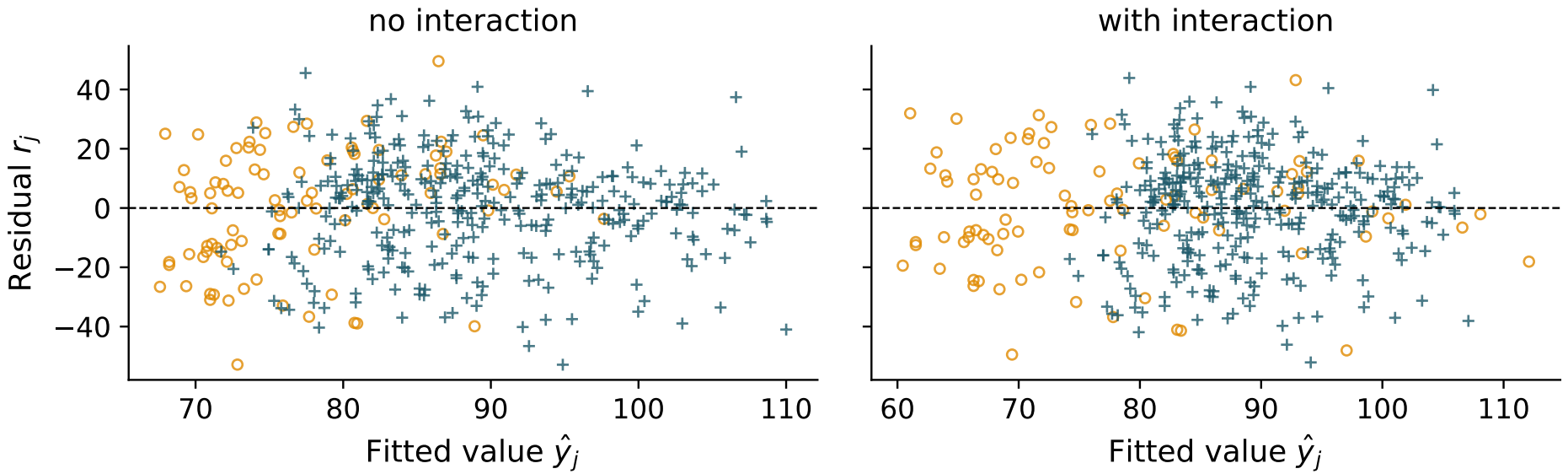




Test scores: residuals vs. fitted values



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Question

The missing interaction is not evident from the residual plot, but it is statistically significant ($p = 0.003$). What does this suggest about how R^2 changes when adding the residual?

Example: advertising budgets and sales

Sales of a product in 200 markets (from *An Introduction to Statistical Learning*): Y is sales in thousands of units, and X_{TV}, X_{radio} are the TV and radio advertising budgets in \$1000s. Without an interaction:

Dep. Variable:		sales	R-squared:			0.897
	coef	std err	t	P>|t|	[0.025	0.975]
const	2.9211	0.294	9.919	0.000	2.340	3.502
TV	0.0458	0.001	32.909	0.000	0.043	0.048
radio	0.1880	0.008	23.382	0.000	0.172	0.204

With an interaction, and both budgets centered at their means (147.0 and 23.3):

Dep. Variable:		sales	R-squared:			0.968
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const	13.9470	0.067	208.737	0.000	13.815	14.079
TV_c	0.0444	0.001	56.673	0.000	0.043	0.046
radio_c	0.1886	0.005	41.806	0.000	0.180	0.198
TV_radio_c	0.0011	5.24e-05	20.727	0.000	0.001	0.001

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How does the effect of an extra \$1000 of radio advertising on sales depend on the TV budget?

Advertising: residuals without vs. with interaction

